HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **AI-native data platform capabilities** to enable self-service data discovery for enterprise users, aligning with the team's mission.
- Built **APIs and services** using **Java Spring Boot** to expose data and business capabilities to AI agents, supporting modern data consumption patterns.
- Contributed to the development of **Retrieval-Augmented Generation (RAG) solutions** by integrating enterprise data sources, enhancing information retrieval.
- Assisted in building and maintaining scalable **data pipelines** and **cloud-native data services** on Azure, improving data processing efficiency.
- Applied engineering best practices including **clean code**, **automated testing**, and **CI/CD** principles to ensure high-quality software delivery.
- Researched and evaluated emerging **AI and data technologies**, assessing their relevance and potential application to the platform's evolution.
- Participated in platform **monitoring, troubleshooting, and continuous improvement** activities, ensuring operational stability and performance.

SKILLS

AI/ML Technologies – AI Agents, Agentic Workflows, Model Context Protocol (MCP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases, Semantic Search

Backend Development – Java, Spring Boot, REST APIs, gRPC

Data Engineering & Processing – Python, Apache Spark, Dremio, Kafka, Data Pipelines, ETL

Cloud Platforms – Azure Data Platform, Databricks, AKS, ADLS, ADF, Azure AI Services

DevOps & Collaboration – GitHub, GitHub Actions, Jira, Confluence, Docker, Kubernetes

Observability & Security – Dynatrace, Datadog, Security, Governance, Data Catalog

PROJECTS

AI Agent Integration for Data Product Discovery

- Developed a **Python-based AI agent** that interfaces with a **vector database** to enable natural language querying of data product metadata.
- Implemented **Retrieval-Augmented Generation (RAG)** patterns to provide contextually relevant data definitions and usage statistics to end-users.
- Designed a **REST API** using **FastAPI** to serve agent responses and integrate with existing platform services, achieving p95 latency under 200ms.

Cloud-Native Data Pipeline for Analytics

- Engineered an **ETL pipeline** using **Python and Apache Spark** on Azure Databricks to process and transform large volumes of structured and unstructured data.
- Built scalable **data services** deployed on **Azure Kubernetes Service (AKS)**, ensuring high availability and efficient data access for downstream analytics.
- Integrated the pipeline with **Azure Data Lake Storage (ADLS)** for robust data storage and management, handling approximately 500 GB of daily data ingestion.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)